Examiner only

5. The release of a cloud of highly toxic methyl isocyanate **(MIC)** gas from a pesticide plant in Bhopal led to the deaths of thousands of people.
On the night of the disaster **MIC** reacted with water that leaked into a storage tank, causing an explosion. As a result, **MIC** was released into the atmosphere.

The chemical formula of MIC is CH_3NCO and the equation for the reaction of MIC with water is shown below.

$$2CH_3NCO + H_2O \rightarrow C_3H_8N_2O + CO_2$$

(a) Calculate the relative molecular mass *(M_r)* of MIC using information from the Periodic Table on page 16. Show your workings, starting with the relative atomic mass (A_r) of each element found in **MIC**. [4]

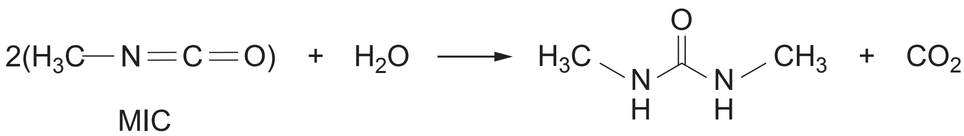
M_r = ...

(b) Approximately 30 000 kg of MIC gas was released. Calculate the number of moles released to three significant figures using your answer in (a). [3]

number of moles = ...

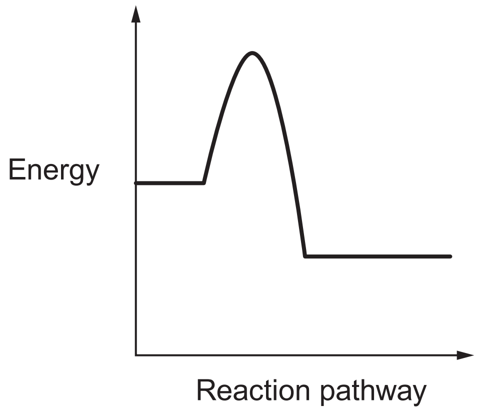
Examiner only

(c) The structural equation and reaction pathway for the reaction causing the explosion in the Bhopal plant are shown below.



MIC

1,3–dimethylurea



Use **all** the information above to explain why the reaction between MIC and water resulted in an explosion in terms of reaction pathway, bond energies and thermal runaway. [6 QER]

13